

Finding the fractions of n*n matrix:

a		b			
9.9	4.4	6.6	7.2	1.5	0.61
0,0	0,1	0,0	0,1		
7.7	1.1	5.5	1.01	1.4	1.08
1,0	1,1	1,0	1,1		

Fractions of n*n matrix

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
void dummy(float a){float *p=&a;}
```

```
void main()
```

```
{
```

```
float a[10][10], b[10][10]; int nr,nc,r,c,rs,cs;
```

```
clrscr();
```

```
printf("Enter no of rows and columns ");
```

```
scanf("%d%d",&nr,&nc);
```

```
printf("Enter %d elements for 1st array  
",nr*nc);
```

```
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%f",&a  
[r][c]);
```

```
printf("Enter %d elements for 2nd array  
",nr*nc);
```

```
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%f",&b  
[r][c]);
```

```
puts("Fractions are ");
```

```
puts("-----");
```

```
for(r=0;r<nr;r++)
```

```
{
```

```
for(c=0;c<nc;c++)
```

```
{
```

```
printf("%10.2f",a[r][c]/b[r][c]);
```

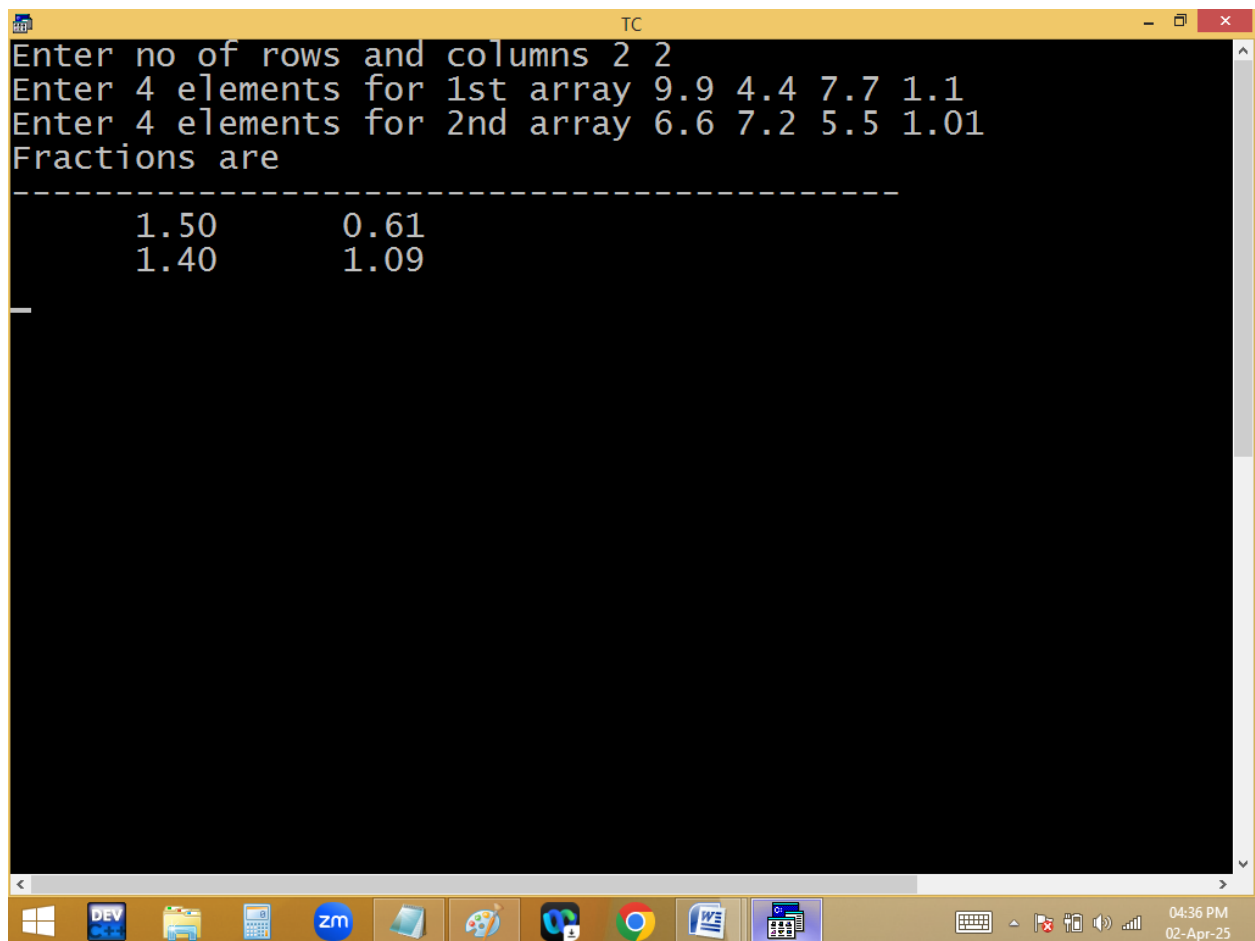
```
}
```

```
printf("\n");
```

```
}
```

```
getch();
```

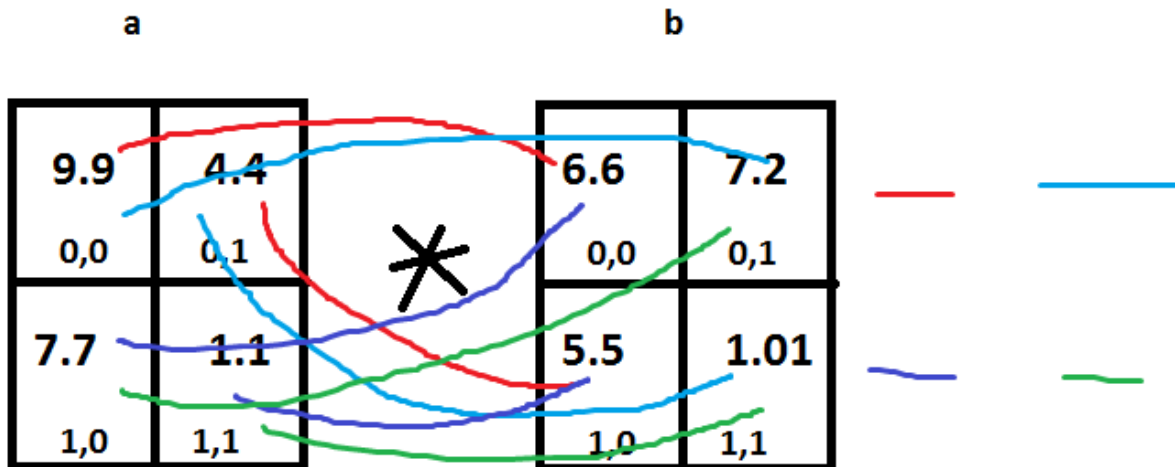
```
}
```



```
TC
Enter no of rows and columns 2 2
Enter 4 elements for 1st array 9.9 4.4 7.7 1.1
Enter 4 elements for 2nd array 6.6 7.2 5.5 1.01
Fractions are
-----
      1.50      0.61
      1.40      1.09
```

```
TC
Enter no of rows and columns 2 3
Enter 6 elements for 1st array 1 2 3 4 5 6
Enter 6 elements for 2nd array 7 8 9 10 11 12
Fractions are
-----
      0.14      0.25      0.33
      0.40      0.45      0.50
```

Matrix Multiplication?



```
#include<stdio.h>
```

```
#include<conio.h>
```

```
void main()
```

```
{
```

```
int a[10][10], b[10][10]; int nr,nc,r,c,rs,cs,s,k;
```

```
clrscr();
```

```
printf("Enter no of rows and columns ");
```

```
scanf("%d%d",&nr,&nc);
```

```
printf("Enter %d elements for 1st array",nr*nc);
```

```
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&
a[r][c]);
```

```
printf("Enter %d elements for 2nd array
",nr*nc);
```

```
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&
b[r][c]);
```

```
puts("Result matrix elements are ");
```

```
puts("-----");
```

```
for(r=0;r<nr;r++)
```

```
{
```

```
for(c=0;c<nc;c++)
```

```
{
```

```
for(s=k=0;k<nc;k++)
```

```
{
```

```
s+=a[r][k]*b[k][c];
```

```
}
```

```
printf("%4d",s);
```

```
}
```

```
printf("\n");
```

```
}
```

```
getch();
```

```
}
```

```

Enter no of rows and columns 2 2
Enter 4 elements for 1st array 1 2 3 4
Enter 4 elements for 2nd array 5 6 7 8
Result matrix elements are
-----
19  22
43  50

```

```

for( r=0; r<2;r++)
{
  for( c=0;c<2;c++)
  {
    for( s=k=0;k<2;k++)
    {
      s+=a[r][k]*b[k][c];
    }
    p(s);
  }
  p("\n");
}

```

1	2
0,0	0,1
3	4
1,0	1,1

5	6
0,0	0,1
7	8
1,0	1,1

$1*5+2*7=19$ $1*6+2*8=22$
 $3*5+4*7=43$ $3*6+4*8=50$

	r	c	k	s
	0	0	0 1 2	$0+1*5+2*7=19$ ✓
	0	1	0 1 2	$0+1*6+2*8=22$ ✓
		2		
	1	0	0 1 2	$0+3*5+4*7=43$ ✓
	1	1	0 1 2	$0+3*6+4*8=50$ ✓

~~2~~

Reading and printing n student data and find the result

```
#include<stdio.h>

#include<conio.h>

void main()

{

int a[10][10], b[10][10]; int nr,nc,r,c,rs,cs,s,k;

clrscr();

printf("Enter no of rows and columns ");

scanf("%d%d",&nr,&nc);

printf("Enter %d elements for 1st array",nr*nc);

for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);

printf("Enter %d elements for 2nd array",nr*nc);
```

```
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&
b[r][c]);
```

```
puts("Result matrix elements are ");
```

```
puts("-----");
```

```
for(r=0;r<nr;r++)
```

```
{
```

```
for(c=0;c<nc;c++)
```

```
{
```

```
for(s=k=0;k<nc;k++)
```

```
{
```

```
s+=a[r][k]*b[k][c];
```

```
}
```

```
printf("%4d",s);
```

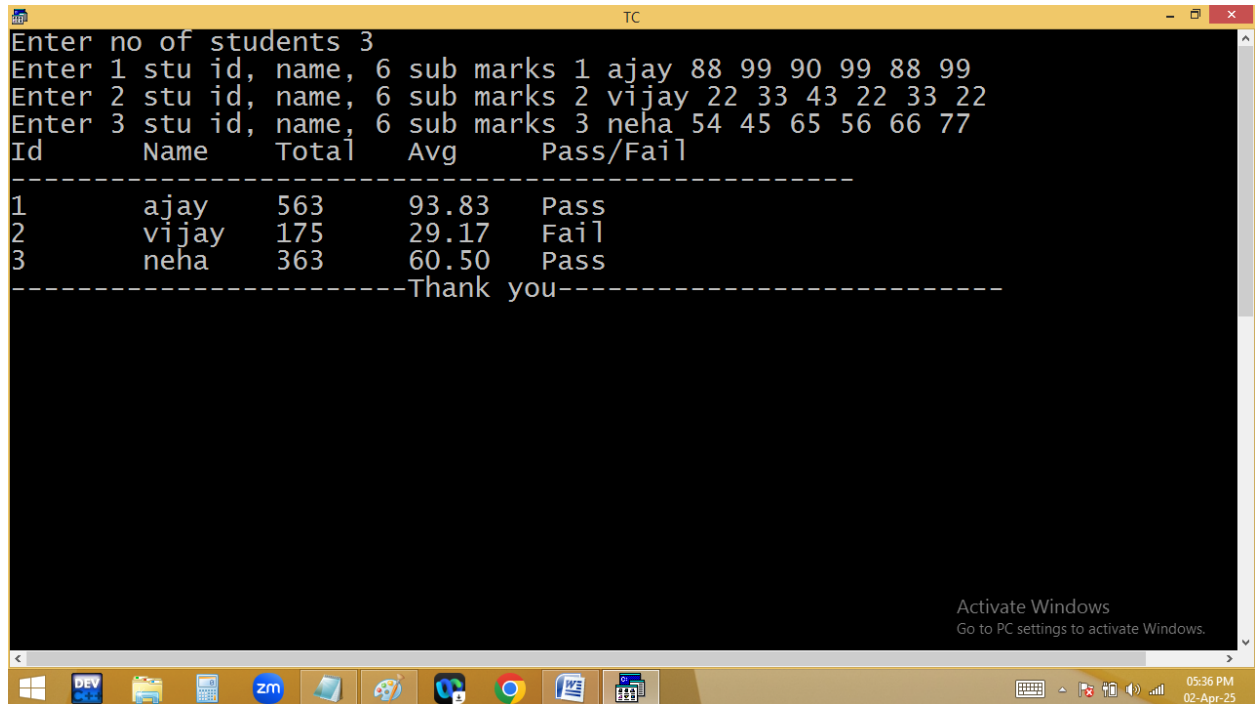
```
}
```

```
printf("\n");
```

}

getch();

}



```
Enter no of students 3
Enter 1 stu id, name, 6 sub marks 1 ajay 88 99 90 99 88 99
Enter 2 stu id, name, 6 sub marks 2 vijay 22 33 43 22 33 22
Enter 3 stu id, name, 6 sub marks 3 neha 54 45 65 56 66 77
Id      Name      Total  Avg      Pass/Fail
-----
1      ajay      563    93.83    Pass
2      vijay      175    29.17    Fail
3      neha      363    60.50    Pass
-----Thank you-----
```

Home work:

Finding stu grade by using above example